

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Java Spring Boot** REST APIs for internal services, handling approximately 500 requests per minute and improving data retrieval efficiency for downstream applications.
- Developed and maintained **Hibernate** ORM mappings and optimized SQL queries for a core enterprise system, reducing average query execution time by **18%** for critical data access operations.
- Wrote comprehensive **JUnit** and **Mockito** tests for new Java features, achieving over **90%** code coverage and ensuring application stability during regression testing.
- Collaborated with FEOL/BEOL Integration and Device teams to analyze structured and unstructured data, identifying root causes for yield excursions and implementing corrective action plans that met committed production targets.
- Analyzed process health data and manufacturing material changes to assess quality and reliability risks, ensuring adherence to Intel's stringent manufacturing standards.
- Established and refined workflows and procedures for continuous improvement programs, preventing yield excursions in line and sort processes across multiple technology nodes.
- Containerized Java applications using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency and reducing manual setup time by **40%**.

SKILLS

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, Hibernate, JUnit, Mockito, pandas, NumPy, scikit-learn, FastAPI

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Technologies – Splunk, Dynatrace, Linux, Jira, Confluence, Power BI, Tableau

Domain Knowledge – Semiconductor Manufacturing, FEOL/BEOL Integration, Yield Analysis, Process Optimization, Data Mining, Statistical Analysis

PROJECTS

Java Application Performance Optimization (HCL Technologies)

- Optimized critical Java Spring Boot microservices by refactoring inefficient code paths and improving database interaction logic, resulting in a **30%** reduction in p95 latency.
- Implemented proactive monitoring using **Dynatrace** and **Splunk** to track application performance and identify bottlenecks, enabling faster root cause analysis for production incidents.
- Integrated a new caching layer using **Redis** within the Java application, reducing direct database load by approximately **20%** during peak traffic hours.

Data Analysis for Process Improvement (MSc Data Analytics)

- Analyzed a dataset of manufacturing process parameters using Python (pandas, NumPy) and R, identifying key factors impacting product yield with statistical significance.
- Developed interactive **Power BI** dashboards to visualize process trends and highlight areas for improvement, presenting actionable insights to academic stakeholders.
- Applied **scikit-learn** to build a predictive model for defect occurrence, achieving an accuracy of **0.82** on a held-out test set.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)